EXPLAINABLE AI · DUKE UNIVERSITY

Which Port Is It Anyway? Rules vs BERT

Auditing how a transparent rule system and a fine-tuned BERT decide which cruise port a review clause is about

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01 · THE PROBLEM

Whose clause is it?

> *"Roatan was stunning but we had a bad experience with a local taxi driver...* ***It left a bit of a sour taste in my mouth.****"*

The bold sentence names no port. Only context says it is about Roatan.

Cruise lines decide which ports and itineraries are worth investing in partly from guest reviews, especially the sentiment in them. But one review moves between ports, the ship and the trip overall, so each clause must be credited to the right place first. If that step is a black box, a cruise line can't tell why a port looks good or bad.

RESEARCH QUESTION

How do a transparent rule system and a fine-tuned BERT decide which destination a clause is about, and are BERT's explanations faithful to what it actually uses?

Contribution: a side-by-side audit that pairs each rule decision trace with BERT's Integrated Gradients attributions, and tests whether those attributions are faithful.

02 · DATA AND METHODS

Two ways to decide

~130 (out of ~600) synthetic reviews (GPT-3) modeled on r/royalcaribbean posts about a Western Caribbean itinerary: Roatán, Honduras; Costa Maya and Cozumel, Mexico. 1,233 annotated clauses, 5 labels (the ports, multiple destinations, and general cruise), split by review; the test set (207 clauses, 20 reviews) is used only once.

Rule system (transparent)

1 · Two ports compared → Multiple Destinations

2 · One port named → that port; opens a 2-sentence window

3 · Shipboard topic → General Cruise; closes the window

4 · Inside the window → inherit the active port

5 · Otherwise → abstain ("Unclear")

Every label comes with a trace: which rule fired, which port, how far back.

Fine-tuned DistilBERT (contextual)

Input: the clause, its sentence, and one sentence before and after

Class-weighted loss, chosen on validation macro-F1 (0.646 vs 0.612 unweighted)

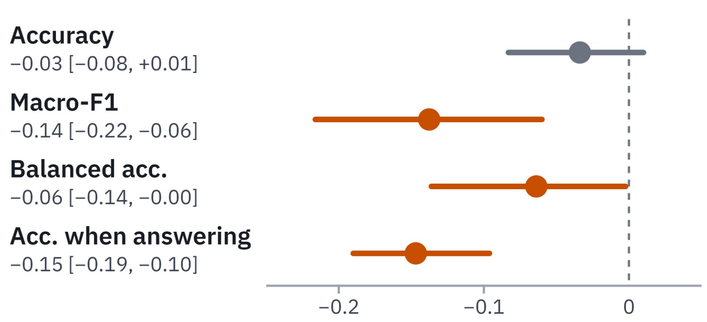
Explained with Integrated Gradients (Captum)

03 · HEADLINE RESULT

Accuracy ties; the rules win on macro-F1

Test set (207 clauses)	Rule	BERT
Accuracy	0.821	0.787
Macro-F1	**0.748**	0.610
Balanced accuracy	0.687	0.624
Coverage (answers given)	0.879	**1.000**
Accuracy when it answers	**0.934**	0.787

BERT – Rule, 95% paired review-bootstrap CI



Left of 0 favors the rule (orange: interval excludes 0). Accuracy's interval crosses 0, so no difference is detectable. The rule's precision is bought by abstaining on 12% of clauses.

Agreement: both correct 149 · rule only 21 · BERT only 14 · both wrong 23

04 · CENTERPIECE: REVIEW 242

One review, opposite failures

Right after "Roatan was beautiful," each method fails exactly where the other succeeds. The rule's topic boundary is correct locally but erases Roatan for every later clause. BERT follows the nearest port name, even across a topic change.

"...but we had a major delay getting off the ship due to local port authorities..."

Gold label: **General Cruise**

Rule — General Cruise · correct

`ship-topic rule fired → active port cleared`

BERT — Roatan · 0.70 · wrong

Top IG tokens: `Roatan +0.78` `ship +0.20`

"We still managed to see the monkeys and sloths,"

Gold label: **Roatan**

Rule — Unclear · abstained

`no active port: it was cleared one clause earlier`

BERT — Roatan · 0.76 · correct

Top IG tokens: `Roatan +0.76` (previous sentence)

"...but everything felt rushed."

Gold label: **Roatan**

Rule — Unclear · abstained

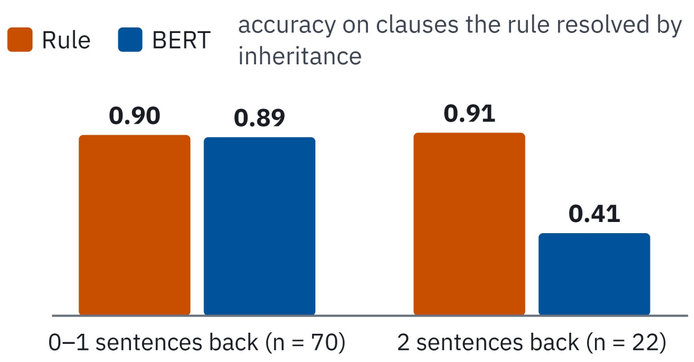
`no active port: the cascade continues`

BERT — Roatan · 0.79 · correct

Same evidence: "Roatan" in the previous sentence

05 · HOW FAR BACK CAN EACH LOOK?

Both follow nearby port names; BERT's reach is shorter



Two sentences back, the port name has left BERT's ±1-sentence input, so it guesses from the clause alone (mean confidence 0.46). This explains 13 of the 21 clauses only the rule got right.

When they agree, it is for the same reason. In the "sour taste" clause (panel 01), the rule inherits Roatan from one sentence back, and BERT's top tokens are "Roatan" in that same sentence (+0.65, +0.64).

06 · HOW THE RULES FAIL

Rules fail loudly, and the trace says why

All 25 rule abstentions trace to four causes:

Ship topic erased the port (cascade)	8
Window expired (3+ sentences)	8
Lead-in before the port is named	6
Generic talk about "the ports"	3

Keyword misfire: "The buffet lunch was cold chicken and rice."

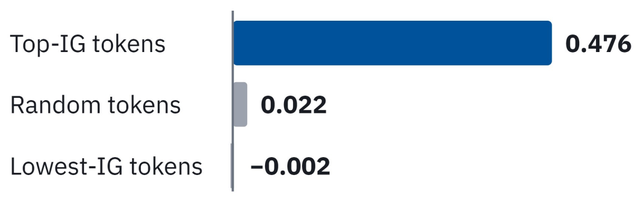
Lunch on a beach excursion (gold: Roatan). The word **`buffet`** fired the ship rule → General Cruise. BERT said Roatan (0.86), led by "Roatan" in the previous sentence.

Wrong, but instantly diagnosable and fixable: the trace names the exact keyword responsible.

07 · CAN WE TRUST THE EXPLANATIONS?

BERT's attributions are faithful

Drop in BERT's predicted-class probability after deleting tokens:



Top – random: +0.45 [0.42, 0.49], all 207 test clauses

What IG reveals: the top token is usually a port-name word piece ("costa", "##mel"), and in 89 of 207 clauses it sits outside the clause itself. BERT is accurate when a port name is in its input (0.82, n = 187) and much less so without one (0.50, n = 20).

Caveat: the punctuation-stability probe changed only 46 inputs (43 kept their label), so it is weak evidence.

08 · BERT READS AHEAD

Looking forward helps, and hurts

"Traveling as a couple, we preferred slower-paced days."

BERT: Cozumel · 0.85 · correct `co ##zu ##mel`

next sentence

"The ports were a mixed bag." (gold: General Cruise)

BERT: Costa Maya · 0.73 · wrong `costa +1.80` `ports −0.40`

Same learned habit, opposite outcomes. With no port in view, BERT defaults to Costa Maya, the largest class: 76 predicted vs 56 gold, and 22 of its 44 errors.

09 · BONUS: DOWNSTREAM SENTIMENT

Do attribution errors change what guests seem to say?

On average, barely. A port's sentiment score (−1 to +1) moves by at most 0.07 under either method. The sentiment scorer (VADER) causes more error than that, up to 0.27.

But two spots need care:

- **Complaints** rarely name the port: 63% of negative comments rely on context to place them, vs 51% of positive ones.
- **Rare categories** swing wildly: of 6 comments comparing ports, the rule caught 1 (score +1.00) and BERT none (no score); guests' real score was +0.33.

10 · LIMITATIONS AND NEXT STEPS

What this does not show

- Small, synthetic corpus. Only 6 Multiple Destinations test clauses (rule 1/6, BERT 0/6)
- Unequal context (rule +2 sentences, BERT ±1) and one BERT seed: this is BERT as configured. IG is post-hoc, not causal.
- Clause boundaries are imperfect: a few segments still join two thoughts, yet carry only one label.

Next: test on a larger set of real guest reviews with more multi-port clauses; give both methods the same 2-sentence lookback and train BERT over several seeds; check IG against a second attribution method; and refine clause splitting before re-labeling.

Integrated Gradients: Sundararajan et al., 2017 · DistilBERT: Sanh et al., 2019 · Captum: Kokhlikyan et al., 2020 · VADER: Hutto & Gilbert, 2014